



T.C.

YILDIZ TEKNİK ÜNİVERSİTESİ

YILDIZ TECHNICAL UNIVERSITY

MEKATRONİK MÜHENDİSLİĞİ BÖLÜMÜ

DEPARTMENT OF MECHATRONICS ENGINEERING

STAJ DEFTERİ

INTERNSHIP NOTEBOOK

ÖĞRENCİNİN / STUDENT'S

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STAJ TÜRÜ / INTERNSHIP TYPE:

☒ Atölye Stajı /Workshop Internship☐ Mesleki Alan Stajı 1/ Occupational Internship 1☐ Mesleki Alan Stajı 2/ Occupational Internship 2

Kurumun Adı/Name of Institution	Bölüm / Department	Çalışma Süresi / Working Time		Hafta/Week
		Tarihinden/From	Tarihine/To	
Adell Armatür ve Vana Fabrikaları A.Ş	Casting	06/07/2026	10/07/2026	1
Adell Armatür ve Vana Fabrikaları A.Ş	Press	13/07/2026	17/07/2026	2
Adell Armatür ve Vana Fabrikaları A.Ş	Mould & tools	20/07/2026	23/07/2026	3
Adell Armatür ve Vana Fabrikaları A.Ş	Machining & Assembly	24/07/2026	24/07/2026	4
Adell Armatür ve Vana Fabrikaları A.Ş	Technical Maintaince	27/07/2026	03/08/2026	4

HAFTALIK ÇALIŞMA / WEEKLY WORK 06/07/2026. 10/07/2026 arası / between			
Gün / Day	YAPILAN İŞLER / ASSIGNED WORKS	Sayfa No / Page Number	Çalışılan Saat / Hours Worked
Pazartesi / Monday	Sand preparation and Mixing Process	4	9
Salı / Tuesday	Core Manufacturing Process	5-6	9
Çarşamba / Wednesday	Core Flash Removal Process	7	9
Perşembe / Thursday	Brass Melting and Pouring Process	8	9
Cuma / Friday	Casting Cooling and Solidification	9	9
Cumartesi / Saturday			
Bölüm / Section :	Casting Department	Toplam Saat / Total Hours:	45
Çalıştığı işyeri ve kısım / Workplace and section : Adell Armatür / Casting Department Kontrol edenin unvanı / Supervisor's title : Adı ve soyadı / Name and surname : Diploma veya oda sicil no / Diploma or chamber registration number :			

HAFTALIK ÇALIŞMA / WEEKLY WORK 13/07/2026. 17/07/2026. arası / between			
Gün / Day	YAPILAN İŞLER / ASSIGNED WORKS	Sayfa No / Page Number	Çalışılan Saat / Hours Worked
Pazartesi / Monday	Introduction to the Hot Pressing Process	10	9
Salı / Tuesday	Automated Hot Pressing Machine	11	9
Çarşamba / Wednesday	Manual Hot Pressing Machine	12-13	9
Perşembe / Thursday	Flash Removal and CNC Machining	14	9
Cuma / Friday	Flash Removal and Final Machining	15	9
Cumartesi / Saturday			
Bölüm / Section :	Press Department	Toplam Saat / Total Hours:	45
Çalıştığı işyeri ve kısım / Workplace and section : Adell Armatür / Press Department Kontrol edenin unvanı / Supervisor's title : Adı ve soyadı / Name and surname : Diploma veya oda sicil no / Diploma or chamber registration number :			

HAFTALIK ÇALIŞMA / WEEKLY WORK 21/07/2026. 27/07/2026. arası / between			
Gün / Day	YAPILAN İŞLER / ASSIGNED WORKS	Sayfa No / Page Number	Çalışılan Saat / Hours Worked
Pazartesi / Monday	Introduction to the Machining Department	16	9
Salı / Tuesday	Cutting Tool Manufacturing	17-18	9
Çarşamba / Wednesday	Tool Inspection and Surface Grinding	19	9
Perşembe / Thursday	Cutting Tools Used in CNC Machining	20	9
Cuma / Friday	CNC Programming and Alignment	21	9
Cumartesi / Saturday			
Bölüm / Section :		Toplam Saat / Total Hours:	45
Çalıştığı işyeri ve kısım / Workplace and section : Adell Armatur / Machining Kontrol edenin unvanı / Supervisor's title : Adı ve soyadı / Name and surname : Diploma veya oda sicil no / Diploma or chamber registration number :			

HAFTALIK ÇALIŞMA / WEEKLY WORK 28/07/2026. 03/08/2026. arası / between			
Gün / Day	YAPILAN İŞLER / ASSIGNED WORKS	Sayfa No / Page Number	Çalışılan Saat / Hours Worked
Pazartesi / Monday	CNC Machining and Final Assembly	21	9
Salı / Tuesday	Final Product Inspection and Assembly	22	9
Çarşamba / Wednesday	Introduction to Programmable Logic Controllers (PLC)	23	9
Perşembe / Thursday	Basic Motor Start EKTS Circuit	24-25	9
Cuma / Friday	Delta and Star Starter Circuit.	26	5
Cumartesi / Saturday			
Bölüm / Section :		Toplam Saat / Total Hours:	41
Çalıştığı işyeri ve kısım / Workplace and section : Assembly & Technical Maintenance Kontrol edenin unvanı / Supervisor's title : Adı ve soyadı / Name and surname : Diploma veya oda sicil no / Diploma or chamber registration number :			

Bölüm / Department:	Machining Industry	Sayfa No / Page No.:	1
Yapılan İş / Assigned Work:	Company Information	Tarih / Date:	06/07/2026
Company Information: Adell Armatür 1.Company History and Profile 1.1History of the Company. Adell Armatür ve Vana Fabrikaları Anonim Şirketi is one of the biggestst manufacturers specializing in sanitary fittings, shoowers systems, and various brass components in Türkiye. Since its foundation in 1981, the company continiously expanded its production scale and invested in numerous engineering technologies thus improving product quality and strengthening its position in domestic and international market stock. A crucial milestone in the company's development was the establishment of Test and Design center, where new products are designed, tested, and continiously improved to satisfy a contantly market deman on a domestic scale as well as on international. Till now Adell Armatur keeps manufacturing products that comply with international quality standards while exporting its products to various market places. 1.2Company Profile Adell Armatür operates in the plumbing fixtures manufacturing industry, specializing in the design and production of sanitary fittings for residential commercial, and industrial applications. The company's product portfolio includes kitchen and bathroom faucets, shower systems, valves, plumbing, fittings, and bathroom accessories. The production process is carred out using advanced manufacturing technologies, including brass casting, hot forging, CNC machining, polishing, electroplating, assembly, and quality control. By integrating these operations withing a single manufacturing facility, Adell keeps maintaining high production efficiency thereby also ensuring the duurability, precision, and reliability of its production. In all its operations, the company emphasizes stable improvement, technological innovation, occupational health and safety, enviromental responsinility, and customer satisfication. These principles have enabled Adell to establish a strong reputation in both the domestic and international markets			
Kontrol Eden Yetkili Ad Soyad / Supervisor's Name Surname:		Şaban Yılmaz	

Bölüm / Department:	OHS standards	Sayfa No / Page No.:	2
Yapılan İş / Assigned Work:	OHS	Tarih / Date:	06/07/2026

2. Occupational Health and Safety

2.1 Importance of Occupational Health and Safety

Occupational health and safety (OHS) is one of the most fundamental principles of a manufacturing industry. Following OHS standards protects employees from workplace hazards prevents occupational accidents and ensures a safe environment for manufacturing production. In heavy manufacturing facilities such as high-temperature, hydraulic systems, or moving mechanical components, strict compliance is required.

2.2 Personal Protective Equipment (PPE)

Employees are required to wear corresponding personal protective equipment according to regulations and potential hazards in the area:

Safety Shoes: Protect feet from falling heavy or sharp objects. Also prevents slipping.

Safety Glasses: Prevent dust, flying particles, or chemical splashes from penetrating eyes.

Protective Gloves: Protect hands from cuts, heat, abrasion, and chemicals.

Safety helmets: Protects head from various impacts.(See Figure 1.)

Ear Plugs: Although required at high noise level, they reduce exposure to excessive noise generated by industrial machinery.(See Figure 2.)

Protective Work Clothing: Protects body from dirt and heat as well as from chemicals.

Protective Masks and Respirators: Prevent the inhalation of dust and airborne particles.



Figure 1: Wearing a safety helmet is compulsory



Figure 2: Wearing ear plugs is mandatory

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Şaban Yılmaz

Bölüm / Department:	Overview	Sayfa No / Page No.:	3
Yapılan İş / Assigned Work:	Manufacturing departments	Tarih / Date:	06/07/2026
3. Manufacturing Departments The manufacturing process at Adell Armatur is diivided into several specialized departments, each having its own subsection and a task. Core Production: Produces sand cores that form the internal cavities of brass castings and ensure a correct and internal geometry of the products. Foundry (Casting): Brass is melted in high-temperature furnaces and then poured into molds cavities containing the sand cores to produce the desired shape of the components. Hot Forging (Pressing): Under huge pressure, workpieces initially heated in a forging furnace are placed uder the hot forging press which then deforms them plastically. CNC Machining: Precise machine operations such as drilling or turning are performed accurately to obtain the required dimensions and surface finish with mimimum tolerance. Surface finishing and Polishing: With latest machine techniques, all surface internals are polished accurately to ensure appropriate dimensions and compliance. Assembly: All components manufactured are then sent to the assembly section and whey are collected into boxes and sent to market destinationos. Quality Control Laboratory: Finished products undergo dimensional insepection, pressure testing, leakage testing, and other quality assessments to ensure compliance with company standards. Packaging and Warehouse: Approved products are packaged, labeled, stored, and prepared for shipment to domestic and international customers. Maintenance: All systems and subsystems are continiously checked and in case of a malfunction, corresponding steps are taken. Research and Development (R&D): New products are designed and text, while exsiting one are continiously improved in accordance to latest engineering techniques. The compliance between all thos departments annsure that manufacturing reaches its high-quality brass sanitary fittings efficiently, while maintaining strict quality control at evert stage of manufactruing at Adell Armatur Furthermore, Adell Armatur also places significant emphasis on occupational health and safety, enviroment consequences, and stable improvement over years. By adopting to worldwide standards, the company secured a highly reputational spot on both domestic and international domestic spot.			
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Bölüm / Department:	Casting	Sayfa No / Page No.:	4
Yapılan İş / Assigned Work:	Sand Preparation and Mixing Process	Tarih / Date:	06/07/2026

Internship: Engineering Journal

1.1 Sand preparation and Mixing Process

Sand Hopper

Stores sand before it is transferred to the same mixing machine.

The team was introduced to the first stages of faucet manufacturing. The first process was the production of the sand core, which is later used in the casting process. Initially, five buckets (see Figure 3) filled with sand are poured into a sand hopper (see Figure 4), from which the sand is discharged into a sand mixer.

Sand Mixing Machine

Mixes sand with resin and hardener to produce a uniform molding mixture.

During the process, each grain gets covered with a microscopic film of resin. Once cured, that thin film bonds millions of grains together. The main purpose of the hardener is to chemically react with the resin thus creating a three-dimensional network. The mixing process must be carefully controlled. If the mixing time is short, the resin and hardener may not react fully, resulting in a non-homogeneous mixture. Otherwise, excessive mixing may cause the mixture to begin curing thus can substantially reduce its effectivity.

Resin Polymerization

The resin and hardener react chemically to bind the sand grains together.

As soon as the resin and hardener come into contact, a chemical reaction begins. During the reaction, small resin molecules join together, these molecules form long polymer chains, and those chains become interconnected into a rigid three-dimensional network. This solid polymer binds the sand grains together. After the mixing process is completed, the prepared mixture is used to produce the sand core. The sand mixing machine used in this process is shown in the Figure 5.



Figure 3



Figure 4



Figure 5

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Bölüm / Department:	Casting	Sayfa No / Page No.:	5
Yapılan İş / Assigned Work:	Core Manufacturing Process	Tarih / Date:	07/07/2026

1.2 Core Manufacturing Process

Parameter	Value
Sand	75 kg
Resin Ratio	1.6 - 2.0 %
Resin	1.2 - 1.5 kg
Hardener	360 - 450 g
Mixing Time	5 min

Table 1: Sand Mixture

Properties	Value
Core Material	Silica Sand
Melting Point	1700 degrees Celsius
Molten Brass	1020 degrees Celsius
Core Box Temperature	200 degrees Celsius

Table 2: Material Properties

Resin Ratio

The resin ratio must be optimized to balance strength, cost, and gas permeability.

Increasing the resin ratio above the optimum does not significantly improve the strength of the core, but increases production cost, reduces gas permeability, generates more smoke during casting, and makes core-removal process difficult.

The mixture ratios are demonstrated in the Table 1.

Core Box

Shapes and cures the sand mixture into the required core geometry.

After the sand mixture is prepared, it is transported in buckets to the core-making machines. The mixture is then blown through tubes into a heated core box maintained at approximately 200 degrees Celsius, where it undergoes curing and takes the required shape. Various Temperatures for the material is illustrated in the Table 2.

Compressed Air System

Uses air pressure to fill the core box uniformly with the sand mixture.

Around 200 degrees is high enough to speed up the chemical reaction between the resin and hardener; fully cure the resin, increasing the mechanical strength of the sand core, and improve the core's strength so it can withstand handling and molten metal. Compressed air is used to blow the sand mixture into the metal core box. The core box has narrow passages and complex cavities. Air pressure distributes sand uniformly thus completely filling the cavity mold.

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Bölüm / Department:	Casting	Sayfa No / Page No.:	6
Yapılan İş / Assigned Work:	Core Manufacturing Process	Tarih / Date:	07/07/2026

Machine Variation

In general, the core production department consists of seven pneumatically driven machines and four hydraulically driven machines. The selection of the appropriate machine primarily depends on the size and dimensional requirements of the workpiece. While pneumatic machine can offer faster and cheaper mass production, hydraulic machine may produce with higher accuracy and much bigger workpieces. The lack of accuracy for pneumatics machine is largely because of air characteristics: air can be easily compressed during the machine operation and thus affect final dimensions of the workpiece. The pressure exerted inside pneumatic machine is estimated to be 6 bar.



On the first picture: Hydraulic machine.

Second picture: Pneumatic machine.

Rest: The internal mechanism of the machines

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Bölüm / Department:	Casting	Sayfa No / Page No.:	7
Yapılan İş / Assigned Work:	Core flash Removal Process	Tarih / Date:	08/07/2026

1.3 Core Flash Removal Process

Flash Removal

Removes excess material from the sand core to ensure dimensional accuracy.

After the cores are produced, excess material (flash) remaining on their surfaces is removed manually using a trimming knife. This ensures that the required dimensions and surface quality. Flash is removed so it is possible to ensure that the core fits accurately inside the mold cavity. Excess flash may reduce dimensional accuracy, cause core misalignment, or produce unwanted fins on the final casting.

Trimming Knife

Used to manually remove flash from finished sand cores.



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Bölüm / Department:	Casting	Sayfa No / Page No.:	8
Yapılan İş / Assigned Work:	Brass Melting and Pouring Process	Tarih / Date:	9/07/2026

1.4 Brass Melting and Pouring Process

Melting Furnace

Melt brass ingots to the required casting temperature.

The mold is then tightly clamped by the casting machine(see Figure 6) to prevent molten metal from leaking during the pouring process. Initially, brass ingots are placed in a furnace heated by natural gas, which raises the temperature to 1020 degrees Celsius, which is sufficient to completely melt the brass alloy used in production.

Pouring System

Transfers molten brass from the furnace to the mold cavity.

Next, with the pressure supplied, compressed air starts transporting brass along through heat-resistant tubes to the pouring nozzle illustrated in the Figure 7 where through a hole brass is poured from the chamber into the mold cavity of the machine. The pouring process lasts approximately 11 seconds. This duration has been determined experimentally to ensure complete filling of the mold while minimizing material waste and preventing casting defects.



Figure 6

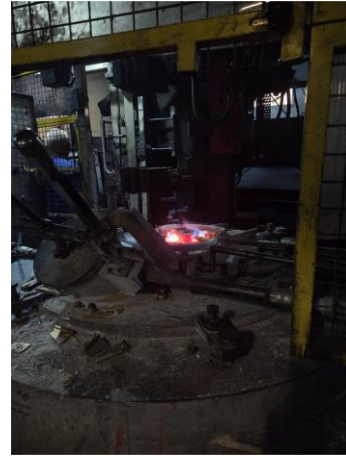


Figure 7

Kontrol Eden Yetkili Ad Soyad / Supervisor's Name Surname:

Şaban Yılmaz

Bölüm / Department:	Casting	Sayfa No / Page No.:	9
Yapılan İş / Assigned Work:	Casting Cooling and Solidification	Tarih / Date:	10/07/2026

1.5 Casting Cooling and Solidification

Solidification

Allows molten brass to harden under controlled conditions inside the mold cavity.

After the mold cavity is completely filled, the molten brass is allowed to solidify before the casting is removed.

Water Cooling Tank

Cools the grippers after casting to prepare them for the next production cycle.

After casting, the grippers holding the cast component at approximately 1000 degrees are immersed in a water-cooling tank to facilitate the next stage of the process. The casting is not cooled immediately because it must solidify under controlled conditions inside the mold. Finally, the casting is released onto the carriage. The step-by-step is shown in the Table 3.

Stage	Process Description	Purpose	Possible Defects if Improperly Handled
Mold Filling	Molten brass is poured into the mold cavity.	Ensures complete filling of the mold.	Misruns, incomplete filling, cold shuts.
Initial Solidification	The outer layer of the molten brass begins to solidify.	Forms the external geometry of the casting.	Surface cracks, poor dimensional accuracy.
Complete Solidification	The remaining molten metal solidifies inside the mold.	Produces a fully solid casting.	Internal shrinkage, porosity, cavities.
Mold Opening	The mold is opened after sufficient solidification.	Allows safe removal of the casting.	Casting deformation if removed too early.
Gripper Cooling	The grippers are immersed in a water-cooling tank.	Reduces the temperature of the handling equipment.	Overheating and reduced equipment life.
Casting release	The casting is placed onto the carriage.	Transfers the product to the next manufacturing stage.	Mechanical damage If handled improperly.
Inspection	The casting is visually inspected before further processing	Detects visible defects before machining.	Defective parts may continue to production

Table 3

Kontrol Eden Yetkili Ad Soyad / Supervisor's Name Surname:	Şaban Yılmaz
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Bölüm / Department:	Press	Sayfa No / Page No.:	10
Yapılan İş / Assigned Work:	Introduction to Hot Pressing Process	Tarih / Date:	14/07/2026

2.1 Introduction to the Hot Pressing Process

Compressor Section

The compressor section manufactures brass components using hot forging technology.

Heating Furnace

The furnace heats brass workpieces to the required forging temperature.

Hot Forging Press

The forging press, which is illustrated in the Figure 3 on the next page, plastically deforms the heated brass into the required shape.

During the second week of the internship, the internship supervisor introduced the team to the compressor section of the factory. The company had been assigned to manufacture a new product using a new machine. Firstly, a small cylindrical part made of brass was heated up to 800 degrees Celsius in a furnace. An indicating signal confirming the readiness of the part for hot pressing is a reddish color. Under careful control, the workpiece was carried from the furnace directly to the forging press, which initially was also heated by coal for better metal flow. When placing the workpiece, the machine was extra lubricated for preventing unexpected outcome, and the part was hot pressed. After validating the results, the work team verified the tolerance of the product.

The characteristics of brass at different forging temperatures are given in the table 4 below:

Temperature in degrees Celsius	Yield Strength(MPa)	Ultimate Tensile Strength(MPa)	Elastic Modulus(GPa)	Thermal Conductivity (W/mK)	Notes
20	180-250	350-500	100-110	110-120	Room temperature
200	120-170	280-400	95-100	115-120	Increased ductility
400	60-100	180-280	85-90	120-125	Suitable for warm forming
600	20-50	80-150	70-80	125-130	Very ductile
700	10-30	40-80	60-70	130-135	Hot forging range begins
800	5-20	20-60	50-60	135-140	Perfect for hot forging
900	2-10	10-40	40-50	140-145	Close to maximum safe limit

Table 4

Kontrol Eden Yetkili Ad Soyad / Supervisor's Name Surname:	Şaban Yılmaz
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Bölüm / Department:	Press	Sayfa No / Page No.:	11
Yapılan İş / Assigned Work:	Automated Hot Pressing Machine	Tarih / Date:	15/07/2026

2.2 Automated Hot Pressing Machine

Automatic Feeding System

Automatically transports brass cylinders from the loading section to the heating unit. The machine is demonstrated in Figure 4.

Electric Drive System

Provides the power required for automatic machine operation.

Production Capacity

Determines the number of products manufactured within a specified period.

Next when confirming that the machine was operating correctly, a second machine was connected. Initially, the machine is loaded with brass cylinders after which it is switched on, and the workpieces are automatically transported to the upper sections for heating. After heating, the workpieces were transferred to the forging press(see Figure 3), where the previously described hot-pressing process took place. The machine is electrically driven. The production capacity of the machine is estimated to be 15,000 pieces per day.

The production of the workpiece is followed by collecting into the carriage which is then transported to the **Mould and Tools** section.



Figure 8: Hydromec - HF 3150 E C/D
hot forging press



Figure 9: Automatic Feeding Machine

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Bölüm / Department:	Press	Sayfa No / Page No.:	12
Yapılan İş / Assigned Work:	Manual Hot Pressing Machine	Tarih / Date:	16/07/2026

2.3 Manual Hot Pressing Machine

Another hot pressing machine had a production capacity of approximately 900 pieces per shift and operated in the same way as the previously-described machine except the part for automatic process where it transports heated workpieces directly to the hot forging machine. The workpiece was heated to 800 degrees Celsius and was put manually inside the cavity mold. After the operator initiated the cycle by pressing a button, an electrically driven mechanism converted the rotary motion of the motor into linear motion thus enabling the press to apply an approximate force of 250,000 N. Under this force, the heated workpiece plastically deformed and was formed into a tubular shape. Some of calculations are presented in the tablow 5 below:

Parameter	Symbol	Formula	Value	Unit	Notes
Shaft Diameter	d	Measured	31	mm	Measured using a vernier caliper.
Shaft Radius	r	d/2	15.5	mm	Used for cross-sectional area calculation.
Cross-setional area	A	$A = \pi \times r^2$	7.55×10^{-4}	m^2	-
Pressing Force	F	Machine Specification	250.000	N	Equal to 250 kN of compressive force.
Total Stroke Length	L	Measured	0.30	m	Totaltravel distance of the hydraulic shaft.
Effective Pressing Distance	s	Measured	0.2	m	Portion of the stroke during which the brass is plastically deformed.
Mechanical Work	W	$W = Fs$	50000	J	Equal to 50 kJ of work performed on the workpiece.
Cycle Time	t	Measured	1	s	Time required to complete a pressing cycle
Average Shaft Speed	v	$V = L/t$	0.3	m/s	Average speed over the complete stroke.
Average Pressing Speed	V_p	$V = s/t$	0.2	m/s	Average speed while brass is compressed
Pressure Applied	P	$P = F / A$	331	MPa	Suitable for hot forging brass

Table 5

Kontrol Eden Yetkili Ad Soyad / Supervisor's Name Surname:	Şaban Yılmaz
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Bölüm / Department:	Press	Sayfa No / Page No.:	13
Yapılan İş / Assigned Work:	Manual Hot Pressing Machine	Tarih / Date:	16/07/2026

The initial workpiece used in the manufacturing process was a brass cylinder. Brass was selected as the material for manufacturing because of its excellent combination of mechanical and physical properties. It possesses high corrosion resistance thereby making it suitable for applications involving prolonged contact with water and humid environments. In addition, brass has good ductility and forgeability. It allows it to be physically deformed at elevated temperatures without cracking. The material also exhibits good machinability thereby enabling subsequent operations such as drilling, threading, and surface finishing to be performed efficiently

Photos of this hot pressing machine are presented below:



Figure 10



Figure 11



Figure 12

Kontrol Eden Yetkili Ad Soyad / Supervisor's Name Surname:

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Bölüm / Department:	Press	Sayfa No / Page No.:	14
Yapılan İş / Assigned Work:	Flash Removal and CNC machining	Tarih / Date:	17/07/2026

2.4 Flash Removal and CNC Machining

Mould and Tools Department

Prepares forged workpieces for the machining process.

Industrial Robot Gripper

Automatically transfers workpieces into the CNC machine.

CNC machine

Removes flash and machines the workpiece to the required dimensions.

In this department, the workpieces are places on a rotating disc, demonstrated in the Figure 8, from which a hand gripper shown in Figure 9 takes the detail and automatically places it inside CNC machine which removes the flash left after hot pressing as demon. Ready details are illustrated below and provided with a technical sketch **on the next page**.



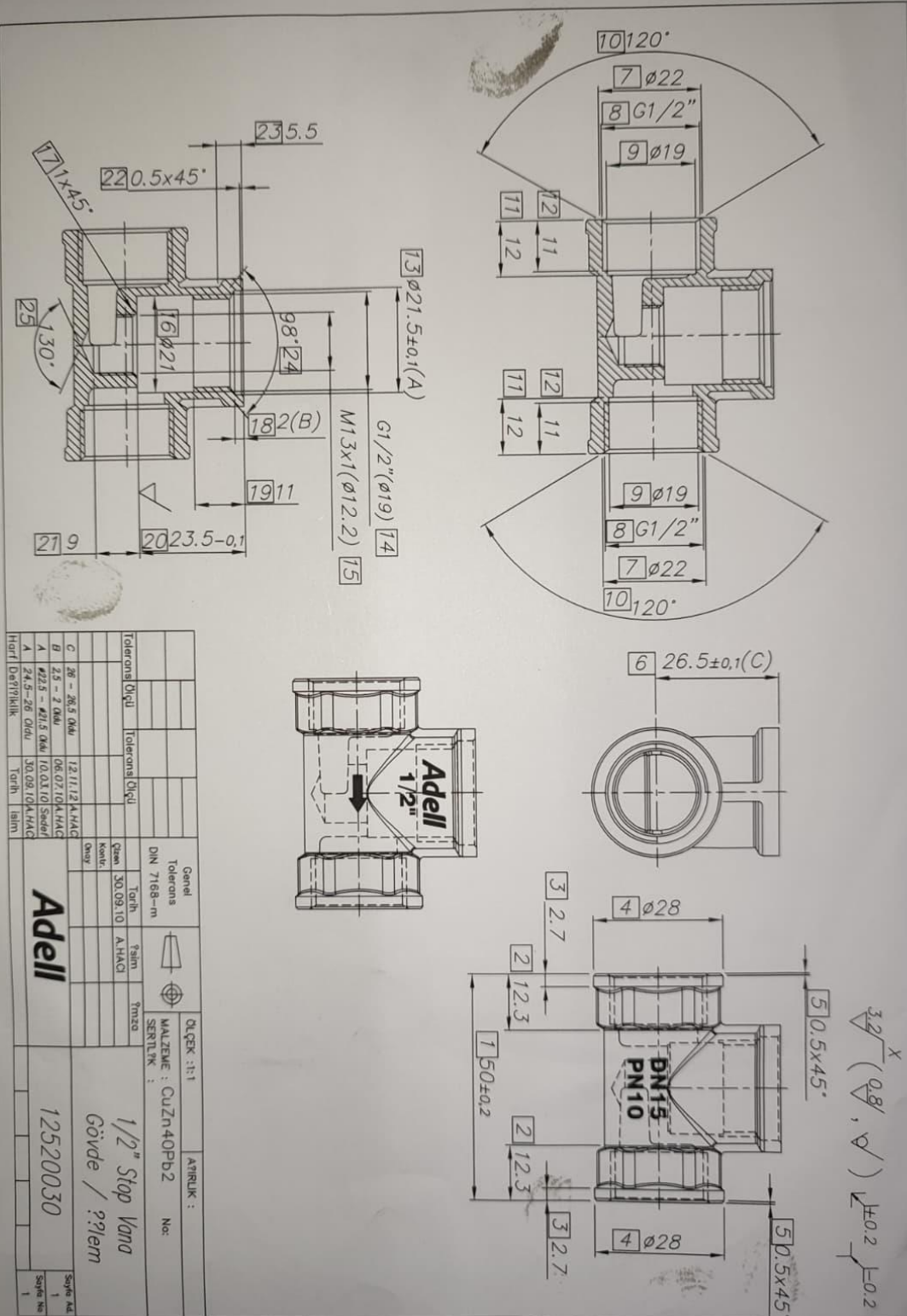
Figure 13



Figure 14

Kontrol Eden Yetkili Ad Soyad / Supervisor's Name Surname:

Şaban Yılmaz



Bölüm / Department:	Press	Sayfa No / Page No.:	15
Yapılan İş / Assigned Work:	Flash removal and Final Machining	Tarih / Date:	20/07/2026

2.5 Flash Removal and Final Machining

Clipper Machines

Remove the flash remaining after the hot forging process.

Pneumatic System

Uses compressed air to operate the flash-removal mechanism.

Hydraulic System

Uses hydraulic pressure to generate high cutting force.

Final Machining

Produces the required dimensions and surface finish before assembly.

With an average production time of approximately 2.6 seconds, the final workpiece was a tube with a flash as seen in the Figure 10, which is then removed by manually operated clipper machines located in another section of the department and also illustrated in the Figure 11. This flash is removed by two machines - one with pneumatic (See Figure 11), and another with hydrolic system (See Figure 12).

After the flash removal process, the forged tubes were transferred to the subsequent production stage for further machining operations. Depending on the product specifications, these operations included machining the inner, and outer surface, drilling holes, or cutting threads to obtain the required dimensions and surface finish. Throughout the manufacturing process, dimensional accuracy and surface quality were periodically inspected to ensure that the components met the company's quality standards before proceeding to assembly or packaging.



Figure 15



Figure 16



Figure 17

Kontrol Eden Yetkili Ad Soyad / Supervisor's Name Surname:

Şaban Yılmaz

Bölüm / Department:	Machining	Sayfa No / Page No.:	16
Yapılan İş / Assigned Work:	Introduction to Machining Department	Tarih / Date:	21/07/2026

3.1 Introduction to the Machining Department

Machining Department

Manufactures mold cavities and performs precision machining operations.

Cavity Mold

Forms the required geometry of the workpiece during forging or casting.

Hot-Work Tool Steel

Provides high hardness and thermal resistance for mold manufacturing.

The third week of the internship was dedicated to the machining section of the factory. In this section a majority of machines consisted of CNC(See Figure 18), milling, grinding, and drilling machines. Specifically here mold cavities are manufactured for the production process.

The cavity mold used in hot forging process or core production were manufactured from steel 2367 and 2344 because of their excellent performance under high-temperature operating conditions. These hot-work tool steels possess high hardness, excellent wear resistance, and good toughness, allowing them to withstand the repeated mechanical loads exerted during forging. In addition, they exhibit high resistance to thermal fatigue and thermal cracking caused by continuous heating and cooling cycles. Their ability to maintain strength and dimensional stability at elevated temperatures increase the service of life of the molds which also ensured consistent product quality throughout the production process.



Figure 18

Kontrol Eden Yetkili Ad Soyad / Supervisor's Name Surname:

Şaban Yılmaz

Bölüm / Department:	Machining	Sayfa No / Page No.:	17
Yapılan İş / Assigned Work:	Cutting Tool Manufacturing	Tarih / Date:	22/07/2026

3.2 Cutting Tool Manufacturing

The first subsection is a place where cutting tools are manufactured. Depending on the required machining operation, different cutting tools were produced, including twist drills, reamers, countersinks, and taps. Each tool possessed a specific geometry designed to achieve the desired cutting performance and dimensional accuracy. After grinding, the tools were inspected to verify their dimensions and cutting-edge quality before being released for use in production.

Cutting Tools

Remove material from the workpiece during machining operations.

Most cutting tools manufactured in this department were made from high-speed steel (HSS). HSS is widely used because it combines high hardness, good toughness, and excellent wear resistance. In comparison to other materials, HSS keeps its material properties at high temperatures during machining which allows higher cutting speed limits and longer tool life. These properties make the material the most preferred among others. Some of the tools are illustrated in the Figure 19 and 20.



Figure 19



Figure 20

Grinding Process

After heat treatment, cutting tools are required to show precision in order to obtain correct dimensions and tolerances. With grinding wheels, a small amount of material left on a workpiece's surface is easily removed thus producing high quality surface finish. Proper grinding also improves cutting performance and reduces tool wear during manufacturing.

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Bölüm / Department:	Machining	Sayfa No / Page No.:	18
Yapılan İş / Assigned Work:	Cutting Tool Manufacturing	Tarih / Date:	22/07/2026

Tool Geometry

The geometry of a cutting tool directly influences cutting process.

Every cutting tool is manufactured with a specific rake angles, clearance angles, and cutting-edge geometry depending on application specifications. Proper geometry reduces cutting forces, improves chip evacuation, and increases the service life. Selecting the correct geometry is essential for achieving dimensional accuracy and surface quality.

Tool Inspection

Manufactured cutting tools are inspected before use in production.

After grinding, each tool is inspected for any defects such as dimensions and cutting quality. With proper dimensions the tools meets specifications, while visual one identifies defects such as flash or blurr left on its surface. Only tools meeting the required specifications are released into the manufacturing production.

The first subsection is a place where cutting tools are manufactured. Depending on the required machining operation, different cutting tools were produced, including twist drills, reamers, countersinks, and taps. Each tool possessed a specific geometry which was designed to achieve the desired cutting performance and dimensional accuracy. Once grinded, the tools were inspected in order to verify their dimensions and cutting-edge quality before being released for use in production.

Property	Typical Value	Unit
Density	8.1	g/cm^3
Hardness (After heat treatment)	62-67	HRC
Tensile Strength	2,500-3,200	MPa
Compressive Strength	3,000-4,000	MPa
Elastic Modulus	210	GPa
Thermal Conductivity	20-30	W/mK
Coefficient of Thermal Expansion	$10.5-11.5 \times 10^{-6}$	$1/^\circ C$
Specific Heat Capacity	460	J/kgK
Melting Temperature	1,400-1,500	$^\circ C$
Maximum Operating Temperature (Hot Hardness)	≈ 600	$^\circ C$

Table 6

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Bölüm / Department:	Machining	Sayfa No / Page No.:	19
Yapılan İş / Assigned Work:	Tool Inspection and Surface Grinding	Tarih / Date:	23/07/2026

3.3 Tool Inspection and Surface Grinding

The optical comparator shown in the Figure 21 was used to inspect the geometry of manufactured cutting tools. After the workpiece was manually clamped onto the machine table, a bright light was projected behind it thereby producing a shadow on a screen. This enlarged profile helped the operator to examine the dimensions, cutting-edge geometry, and possible defects such as burrs or profile deviations. If any imperfections were detected, the workpiece was transferred to a surface polishing machine for further finishing.



Figure 21



Figure 22

The surface-polishing machine is illustrated in the Figure 22. Initially a workpiece is secured on a magnetic chuck. Next after pressing a button, an operator manually adjusts the speed and the starting position of the grinding wheel. Due to high friction generated between the grinding wheel and the workpiece, the workpiece is constantly supplied with coolant during the process to minimize thermal effect on the workpiece and the disk. After completing the process, the surface is cleaned and edges of the detail are manually rounded to provide better stress concentration during the subsequent manufacturing stage. The initial and final workpiece after the polishing process is illustrated in the Figures 23 and 24.



Figure 23



Figure 24

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Bölüm / Department:	Machining	Sayfa No / Page No.:	20
Yapılan İş / Assigned Work:	Cutting Tools Used in CNC Machining	Tarih / Date:	26/07/2026

3.4 Cutting Tools Used in CNC Machining

There is a variety of cutting tools, and each of them serves its own purpose which is illustrated in Table 7.

Tool	Application and Purpose
End Mill (Figure 25)	Milling flat surfaces, slots, pockets, contours, and profiles.
Face Mill (Figure 26)	Machining large flat surfaces with a high surface finish.
Twist Drill	Producing cylindrical holes.
Reamer (Figure 27)	Finishing drilled holes to achieve precise diameter and smooth surface finish.
Tap	Cutting internal threads.
Boring Bar	Enlarging and accurately finishing existing holes.
Ball Nose End Mill	Machining 3D surfaces, molds, and dies.
Chamfer Mill	Creating chamfers, deburring edges, and countersinking holes.
Turning Tool	External turning, facing, and profiling of cylindrical parts.
Grooving / Parting Tool	Cutting grooves and separating finished parts from stock.



Figure 25



Figure 26



Figure 27

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Yapılan İş / Assigned Work:	CNC Programming and Alignment	Tarih / Date:	27/07/2026

3.5 CNC Programming and Alignment

After selecting a proper cutting tool, the next manufacturing stage is CNC machining. With a cutting path designed in software called ESPRIT EDGE, the code is loaded in a chip which is then inserted in a CNC controller.

However, before the program was loaded into the CNC machine, the tool path was prepared in ESPRIT EDGE. In this software, the operator selected the cutting tools, defined machining process. The simulation helped to detect possible collisions and verify that the machining scheme was correct before the manufacturing production officially starts.

Next, a dial indicator illustrated in the Figure 28 below is used for precise alignment and zero setting. It also measures runout of a tool holder, spindle, or cylindrical part. Initially, the dial indicator is mounted on a workpiece: it is important that the contaminants do not influence measurement accuracy. Then the contact tip touches the workpiece surface. As the machine moves, any deviation causes the needle to move. The operator adjusts the workpiece position until the reading reaches the required value.



Figure 28



Figure 29

However, the method mentioned is applicable for altitude only. Another method zero setting is called Edge Finding. The primary objective of this method is to set X and Y axis zero. The main tool used in this method is called edge finder. There are several types of edge finders depending on their applications, but the principle remains the same. A mechanical type is the cheapest and most effective tool making it the most preferred one. It has a spring-loaded tip and as the rotating tip touches the working edge, it suddenly kicks or slides off-center, indicating the exact edge. The tool is illustrated in the Figure 29.

After the workpiece coordinate system is established, the machining program is started. Before beginning the actual cutting process, the operator usually performs a dry run or checks the tool path on the machine to ensure that no collisions will occur. This step helps to prevent damage to the cutting tool, workpiece, and machine.

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Bölüm / Department:	Machining	Sayfa No / Page No.:	22
Yapılan İş / Assigned Work:	CNC Machining and Final Assembly	Tarih / Date:	28/07/2026

4.1 CNC Machining and Final Assembly

CNC machining is one of the most curical steps in the manufacturing production since it ensures that every product meets required specifications regarding the dimensional accuracy and surface quality. Despite the fact that brass workpieces are already close to their final shape after forging, small amounts of material are removed to achieve required tolerances. This stage also provides a smoother surface finish, allowing the components to fit correctly during assembly thus ensuring proper and reliable performance in the final product before distributions.

After he flash removal process, products are transported to the CNC machine sections. With the same operating principle as with mold cavities, CNC machines are programmed with a CNC code with which they produce the final dimensions and surface finish.

Firstly, workpieces are inspected manually for defects such as unremoved flash or incorrect dimensions. Once validated, they are placed inside the CNC machine. By pressing the start button, a workpiece is clamped by two pneumatic-driven metallic plates. Once the workpiece is secured, doors of the machine are automatically closed and using the CNC program, the machine starts the polishing process the workpiece. Upon the completion of the process, the final stage is to check threads and dimensions. Dimensions are checked with a vernier calpier, whereas the threads by thread plug gauge and thread ring gauge.

After the machining operations were completed, the products were transferred to the assembly section, where they underwent further assembly and final inspection before being packaged and prepared for distribution. The process is illusrated in the Figures 30 and 31.



Figure 30



Figure 31

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Bölüm / Department:	Assembly	Sayfa No / Page No.:	23
Yapılan İş / Assigned Work:	Final Product Inspection and Assembly	Tarih / Date:	29/07/2026

4.2 Final Product Inspection and Assembly

Pressure testing is a critical quality control step in a faucet manufacturing since it ensures that the final product safely withstands operating conditions thus meeting all the requirements and standards. The test verifies the reliability of internal channels, connections, and sealing components before the products are transported to the next manufacturing stage. Once defective products are identified, the company is able to prevent failures during customer use. This process demonstrates the importance of automated testing systems and inspection procedures.

The testing process initially begins with firstly putting the faucet with 2 apertures into the testing fixture, where two tubes were connected to. The pneumatic system, shown in the Figure 32, supplied compressed air into the faucet, thereby increasing the pressure. According to the applicable standards, the pressure was set to 3 bar at the outlet of the faucet, and 6 bars at the inlet. If the pressure was observed to drop during the test, then the faucet was considered defective and was rejected for disposal.



Figure 32

In case of absence of defects, the product is carried out to the next stage where it is assembled into boxes and sent to distributors.

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Bölüm / Department:	Technical Maintenance	Sayfa No / Page No.:	24
Yapılan İş / Assigned Work:	Introduction to PLC	Tarih / Date:	30/07/2026

4.3 Introduction to Programmable Logic Controllers (PLC)

A PLC, or Programmable Logic Controller, is a computer designed to operate in harsh environments such as extreme temperatures, dusty areas, or fields outside of human capabilities. PLCs are aimed to handle tasks such as automation of mechanical tasks in industrial factories or manufacturing plants. It also significantly changes operational needs thus enhancing adaptability in industrial environments.

The main advantage of PLCs over wired control systems is the ability to be easily reprogrammed once environments of the manufacturing process change. In control systems, a change in the environment leads to time-consuming wire unplugging in order to change the system. However, in PLC-based machines it only takes operator's time to tune machine according to the working conditions.

PLCs are widely used in manufacturing because they increase production efficiency, improve product quality, and reduce the need for manual operation. They also make it easier to modify or expand automated systems by simply changing the control program instead of rewriting every electrical circuit.

Before using PLCs, many industrial control systems strongly relied on the systems based on relay logic. Conventional relay-based control systems provide the fundamental principles of industrial automation. Prior knowledge acquired through **EKTS** (Electrical Control Techniques Simulator). It is widely used in engineering and vocational education to teach the principles of industrial electrical control before working with real PLC-based systems. One of the simplest EKTS circuits is presented in the Figure 33.

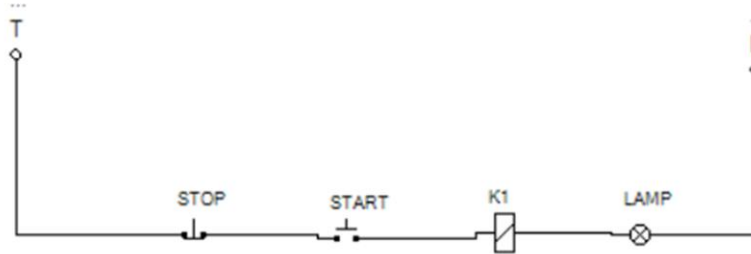


Figure 33

The Figure 33 demonstrates a simple lamp circuit. It has **STOP button**, and **START button** as well as the **Contactor** named K1 and a **lamp**. Once the START button is pressed, K1 is energized, and, consequently, the lamp turns on. However, once the START button is released, K1 loses the connection with the energy source and the whole system goes off. This makes the STOP button particularly irrelevant in this application.

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In order to solve the problem detect in the circuit demonstrated in the Figure 31, a normally opened contact is implemented as illustrated in the Figure 34.

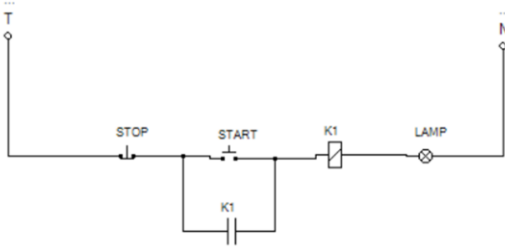


Figure 34

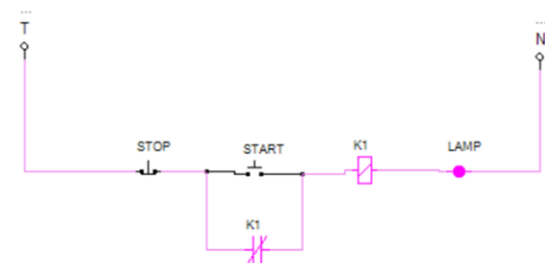


Figure 35

Normally opened contact is not connected. This makes the circuit 34 specifically practical. Once the START button is pressed, the contactor K1 is energized, and K1 contact becomes **normally closed contact** thus commencing the energy transfer. In comparison to the previous circuit in the Figure 33, once the START button is released, the circuit keeps operatins since K1 contact creates an alternative current path that bypasses the START button, and, consequently, constantly supplying the lamp with electricity. The operating circuit principle is clearly shown in the Figure 34.

The STOP button is implemented as a normally closed contact to ensure safe operation. Under normal conditions, it allows current to flow to the contactor coil. Onc the STOP button is pressed, the circuit is opened and thus it immediately de-energizes the contactor and disconnects the load. This configuration is commonly adopted in industrial control systems because it enables rapid shutdown of equipment during normal operation or emergency situation.

This type of control circuit is widely used in industrial automation for controlling electric motors, conveyor systems, pumps, and other electrically driven equipment. The holding circuit prevents the operator from continously pressing the START button during operation, thereby improving convinience and ensuring reliable machine control.

The final result of the circuit is shown in the Figure 36.

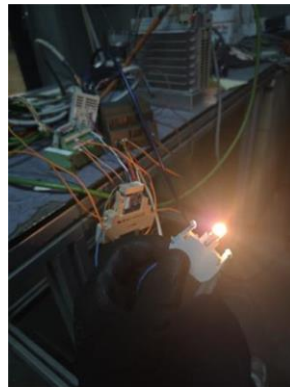


Figure 36

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Yapılan İş / <i>Assigned Work:</i>	Basic Motor Start EKTS Circuit	Tarih / <i>Date:</i>	03/08/2026

4.4 Basic Motor Start EKTS Circuit

Another example of EKTS circuit is a motor starter circuit. In this circuit, a real-world application of EKTS is demonstrated in the Figure 37.

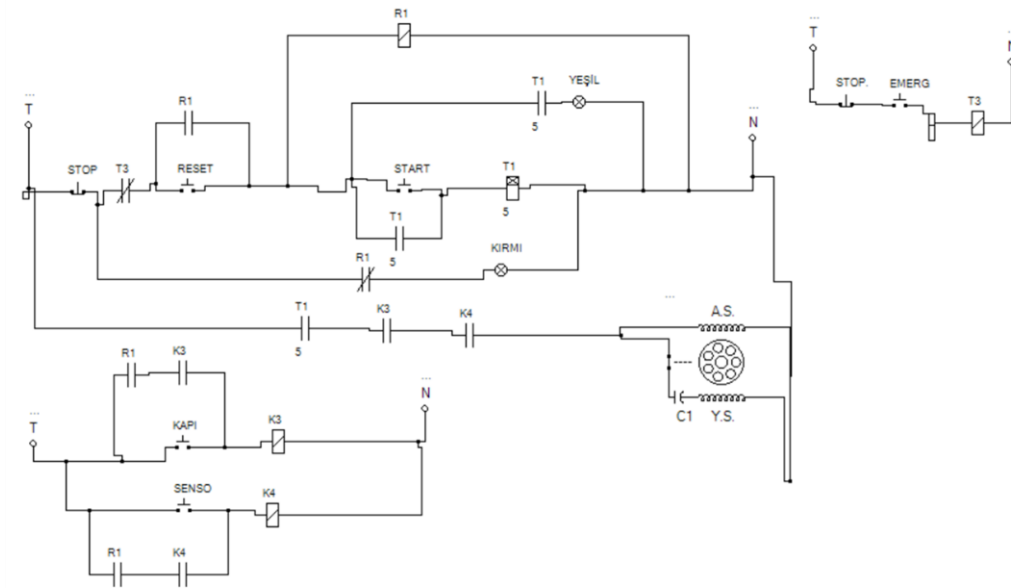


Figure 37

Firstly, the circuit consists of 3 subcircuits: **main, control, and emergency.**

Main. In addition to START and STOP buttons, a **RESET** button is added. Before any buttons are pressed, the lamp called “Kırmı” starts blighting with red color, indicating that the machine is inactive. Later, a on-relay is implemented. In order for the main circuit to turn on, the START button has to be presed for 5 seconds to activate the machine thus turning the green lamp on, and turning off the red one.

Control. However, for the motor to start operating, two conditions have to be met. The first one is checking the door state. If doors are opened, the machine cannot be started. Another condition is a sensor check. Once both of those buttons are pressed, the machine starts operating.

Emergency. In every industrial machine the presence of an emergency button is a mandatory standard. In this subcircuit, the components represent basic START/STOP scheme. Once the **EMERG** button is pressed, T3 contactor is energized, and T3 normally-closed contact which is located in the main circuit between STOP and START button becomes **normally-opened** thus cutting the energy connection. In order for the machine to start operating again, the whole procedure should be repeated.

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Bölüm / Department:	Technical Maintenance	Sayfa No / Page No.:	27
Yapılan İş / Assigned Work:	Delta and Star Starter Circuit	Tarih / Date:	07/07/2026

4.5 Delta and Star Starter Circuit.

A star-delta starter is an electromechanical starting device that uses contactors and a timer to change the winding configuration of a three-phase induction motor from star to delta.

When a three-phase induction motor starts directly at full voltage, it draws 5 to 8 times its rated full-load current. This high inrush current causes voltage dips in the supply network affecting the overall equipment, excessive heating in the motor windings, huge mechanical stress on the shaft, coupling, and driven load, and tipping of protective devices.

For motors above 5 HP (3.7 kW), most electricity boards mandate reduced-voltage starting. The star-delta achieves this without any resistors, autotransformers, or any other electrical components.

The selection between star connection and star-delta starting depends mainly on power ratings of the motor. A star connection is generally used for smaller motors or, for example, for motors designed to operate continuously in a star configuration. In this connection, each motor receives a lower voltage compared to the line voltage. However, since the applied voltage is reduced, the torque will be smaller as well. This torque reduction makes this configuration unsuitable for applications requiring high torque. The formula for the voltage reduction in a star configuration is given below:

$$V_{phase} = \frac{V_{line}}{\sqrt{3}}$$

Star-delta starting is commonly applied to a larger three-phase induction motor where direct starting would result in an excessive current demand. Once started, the motor is initially connected to a star configuration thereby reducing the voltage applied to each winding and limiting the starting current. After the motor reaches a sufficient speed (about 90 %), the circuit is automatically switched to a delta configuration, allowing the motor to operate at its rated voltage and power. This method reduces the voltage drops in the electrical network and provides smoother starting of high power equipment.

The EKTS circuit of star-delta starter is presented in the Figure 38. The motor operates 5 seconds in a star configuration (the timeline is selected randomly) and then automatically switches to a delta connection.

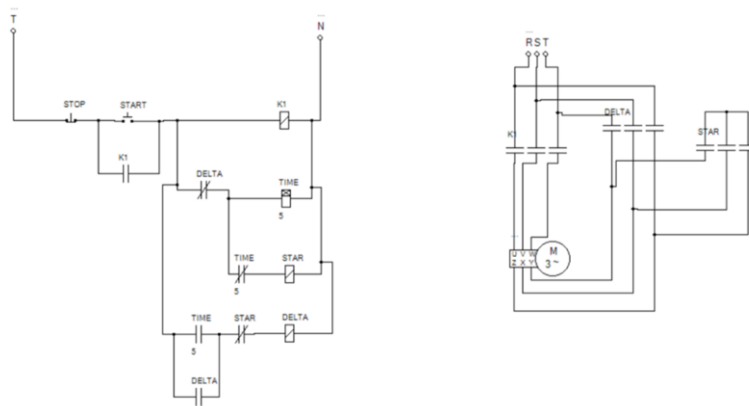


Figure 38

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